

Linux Commands with macOS Versions

Clean study note with practical examples. Many command names are the same on both systems, but macOS often uses BSD tool variants while Linux commonly uses GNU variants. That means the basic command may match, but some options can differ.

Study tips: learn the command purpose first, then run the example, then change the file or folder names to make the command feel natural.

What to remember	Quick note
Same command, different flags	Common with sed, date, stat, and ps.
Linux package managers	apt, dnf, yum, pacman depend on the distribution.
macOS package manager	Homebrew is the usual choice: brew install <package>.
Linux-only tools	Examples: ip, lsblk, free, systemctl.
macOS alternatives	Examples: ifconfig, diskutil, vm_stat, launchctl, open.

1. Navigation and File Basics

Command	Purpose	macOS version / note	Example
pwd	Print the current working directory.	Same on macOS	pwd
ls	List files and folders.	Same on macOS	ls
ls -la	List detailed entries including hidden files.	Same on macOS	ls -la
cd /path	Move into another directory.	Same on macOS	cd /Users/inz/Documents
cd ..	Go to the parent directory.	Same on macOS	cd ..
mkdir notes	Create a new directory.	Same on macOS	mkdir os-notes
rmdir notes	Remove an empty directory.	Same on macOS	rmdir old-notes
touch file.txt	Create an empty file or update timestamp.	Same on macOS	touch commands.txt
cp a b	Copy a file.	Same on macOS	cp notes.txt backup.txt
cp -r src dst	Copy a directory recursively.	Same on macOS	cp -r unit1 unit1-copy
mv a b	Move or rename a file or directory.	Same on macOS	mv draft.txt final.txt
rm file	Delete a file.	Same on macOS	rm temp.txt
rm -r dir	Delete a directory and its contents.	Same on macOS	rm -r old-project
ln -s target link	Create a symbolic link.	Same on macOS	ln -s /usr/bin/python3 py3
file name	Show the file type.	Same on macOS	file report.pdf

2. Viewing and Editing Text

Command	Purpose	macOS version / note	Example
cat file	Print the whole file.	Same on macOS	cat notes.txt
less file	View a file page by page.	Same on macOS	less system.log
head file	Show the first 10 lines.	Same on macOS	head notes.txt
head -n 20 file	Show the first 20 lines.	Same on macOS	head -n 20 log.txt
tail file	Show the last 10 lines.	Same on macOS	tail notes.txt
tail -f file	Follow a file as new lines are added.	Same on macOS	tail -f app.log
nano file	Edit a file in the Nano editor.	Same on macOS	nano todo.txt
vim file	Edit a file in Vim.	Same on macOS	vim config.txt
wc file	Count lines, words, and bytes.	Same on macOS	wc notes.txt
sort file	Sort lines alphabetically.	Same on macOS	sort names.txt
uniq file	Remove adjacent duplicate lines.	Same on macOS	uniq sorted.txt
cut -d: -f1 file	Extract fields from each line.	Same on macOS	cut -d: -f1 /etc/passwd
paste a b	Merge lines from files side by side.	Same on macOS	paste names.txt ids.txt
tr a-z A-Z	Translate characters.	Same on macOS	echo linux tr a-z A-Z

3. Searching and Filters

Command	Purpose	macOS version / note	Example
grep word file	Search for matching text.	Same on macOS	grep kernel notes.txt
grep -i word file	Search without case sensitivity.	Same on macOS	grep -i error app.log
grep -r word dir	Search recursively in folders.	Same on macOS	grep -r scheduler ./notes
find . -name '*.c'	Find files by name pattern.	Same on macOS	find . -name '*.txt'
find . -type d	Find only directories.	Same on macOS	find . -type d
which cmd	Show the path of a command.	Same on macOS	which ssh
command -v cmd	Reliable way to check whether a command exists.	Exists on macOS	command -v git
locate name	Search the system file index.	Same if configured	locate hosts
diff a b	Compare two files line by line.	Same on macOS	diff old.txt new.txt
sed 's/a/b/' file	Replace the first match in each line.	BSD sed on macOS; syntax differs from GNU sed	sed -i 's/Linux/macOS/' note.txt
awk '{print \$1}' file	Process text by fields.	Same on macOS	awk '{print \$1}' marks.txt
rg term	Fast recursive search with ripgrep.	Available via Homebrew on macOS	rg semaphore

4. Permissions and Ownership

Command	Purpose	macOS version / note	Example
chmod 644 file	Set read and write permissions.	Same on macOS	chmod 644 notes.txt
chmod +x script.sh	Make a file executable.	Same on macOS	chmod +x run.sh
chown user file	Change file owner.	Same on macOS	sudo chown inz notes.txt
chgrp staff file	Change file group.	Same on macOS	chgrp staff notes.txt
umask	Show the default permission mask.	Same on macOS	umask
id	Show user ID and group IDs.	Same on macOS	id
groups	Show your groups.	Same on macOS	groups

5. Processes and Jobs

Command	Purpose	macOS version / note	Example
ps	Show running processes for the current shell.	Same on macOS	ps
ps aux	Show a detailed process list.	Same on macOS	ps aux
top	Interactive process viewer.	Same on macOS	top
kill PID	Send a signal to a process.	Same on macOS	kill 4210
kill -9 PID	Force kill a process.	Same on macOS	kill -9 4210
pkill name	Kill a process by name.	Same on macOS	pkill node
jobs	List background jobs started from the shell.	Same on macOS	jobs
bg	Resume a stopped job in the background.	Same on macOS	bg

Command	Purpose	macOS version / note	Example
fg	Bring a background job to the foreground.	Same on macOS	fg
command &	Run a command in the background.	Same on macOS	python3 server.py &
nohup command &	Run a command that survives terminal closure.	Same on macOS	nohup long_task.sh &

6. System Information

Command	Purpose	macOS version / note	Example
uname -a	Show kernel and machine details.	Same on macOS	uname -a
hostname	Show the system host name.	Same on macOS	hostname
whoami	Show the current username.	Same on macOS	whoami
who	Show logged-in users.	Same on macOS	who
uptime	Show how long the system has been running.	Same on macOS	uptime
date	Show or format the date and time.	Same name, flags differ	date
cal	Display a calendar.	Same on macOS	cal
env	List environment variables.	Same on macOS	env
export VAR=value	Create or update an environment variable.	Same on macOS	export EDITOR=vim
free -h	Show memory usage.	Linux only	Use vm_stat on macOS
vm_stat	Show virtual memory statistics.	macOS alternative	vm_stat
sw_vers	Show the macOS version.	macOS only	sw_vers
system_profiler	Detailed hardware and software report.	macOS only	system_profiler SPHardwareDataType

7. Disks and Storage

Command	Purpose	macOS version / note	Example
df -h	Show free disk space in human-readable form.	Same on macOS	df -h
du -sh dir	Show the size of a directory.	Same on macOS	du -sh Downloads
mount	List mounted file systems.	Same on macOS	mount
umount path	Unmount a file system.	Same spelling on macOS; diskutil is common	umount /mnt/usb
lsblk	List block devices.	Linux only	Use diskutil list on macOS
diskutil list	List disks and partitions.	macOS alternative	diskutil list
fsck	Check and repair a file system.	Same on macOS	sudo fsck /dev/disk2s1

8. Networking

Command	Purpose	macOS version / note	Example
ip addr	Show network interfaces and IP addresses.	Linux only	Use ifconfig on macOS
ifconfig	Show or configure interfaces.	Present on macOS; older but common	ifconfig

Command	Purpose	macOS version / note	Example
ping host	Test connectivity to a host.	Same on macOS	ping google.com
netstat -rn	Show the routing table.	Works on macOS	netstat -rn
route -n get default	Show the default route.	macOS common form	route -n get default
dig domain	Query DNS records.	Same on macOS	dig openai.com
nslookup domain	Simple DNS lookup.	Same on macOS	nslookup example.com
nc -vz host port	Test whether a TCP port is open.	Same on macOS	nc -vz localhost 22
curl URL	Fetch data from a URL.	Same on macOS	curl https://example.com
wget URL	Download a file from the web.	Often default on Linux; Homebrew on macOS	wget https://example.com/file.txt
ssh user@host	Log in to a remote machine securely.	Same on macOS	ssh user@192.168.1.20
scp file user@host:path	Securely copy a file to another machine.	Same on macOS	scp notes.txt user@server:/tmp/
rsync -av src dst	Sync files efficiently.	Same on macOS	rsync -av project/ backup/
lsof -i :3000	Show which process uses a network port.	Useful on macOS	lsof -i :3000
traceroute host	Trace the network path to a host.	Same on macOS	traceroute example.com

9. Archives and Compression

Command	Purpose	macOS version / note	Example
tar -cvf file.tar dir	Create a tar archive.	Same on macOS	tar -cvf notes.tar notes/
tar -xvf file.tar	Extract a tar archive.	Same on macOS	tar -xvf notes.tar
tar -czvf file.tar.gz dir	Create a gzip-compressed tar archive.	Same on macOS	tar -czvf src.tar.gz src/
gzip file	Compress a file with gzip.	Same on macOS	gzip report.txt
gunzip file.gz	Uncompress a gzip file.	Same on macOS	gunzip report.txt.gz
zip -r file.zip dir	Create a zip archive.	Same on macOS	zip -r notes.zip notes/
unzip file.zip	Extract a zip archive.	Same on macOS	unzip notes.zip

10. Users and Privileges

Command	Purpose	macOS version / note	Example
sudo command	Run a command with elevated privileges.	Same on macOS	sudo ls /var/root
su user	Switch to another user account.	Same on macOS	su root
passwd	Change your password.	Same on macOS	passwd
useradd name	Create a new user.	Linux common command	sudo useradd student
sysadminctl -addUser name	Create a user on macOS.	macOS alternative	sudo sysadminctl -addUser student
dscl . -list /Users	List local users on macOS.	macOS specific	dscl . -list /Users

11. Package Management

Command	Purpose	macOS version / note	Example
apt update	Refresh package lists on Debian or Ubuntu.	No direct macOS equivalent	sudo apt update
apt install pkg	Install a package on Debian or Ubuntu.	No direct macOS equivalent	sudo apt install git
dnf install pkg	Install a package on Fedora.	No direct macOS equivalent	sudo dnf install git
yum install pkg	Install a package on older RHEL systems.	No direct macOS equivalent	sudo yum install git
pacman -S pkg	Install a package on Arch Linux.	No direct macOS equivalent	sudo pacman -S git
brew install pkg	Install a package with Homebrew.	macOS standard choice	brew install wget
brew search pkg	Search Homebrew packages.	macOS standard choice	brew search python
brew upgrade	Upgrade installed Homebrew packages.	macOS standard choice	brew upgrade

12. macOS-Specific Commands You Should Know

Command	Purpose	macOS version / note	Example
open file	Open a file or app from Terminal.	macOS only	open .
pbcopy	Copy stdin into the clipboard.	macOS only	cat notes.txt pbcopy
pbpaste	Paste clipboard contents to stdout.	macOS only	pbpaste
mdfind query	Search with Spotlight metadata.	macOS only	mdfind kernel
mdls file	Show Spotlight metadata for a file.	macOS only	mdls report.pdf
defaults read domain	Read macOS preference settings.	macOS only	defaults read NSGlobalDomain
launchctl list	Show launch agents and daemons.	macOS replacement for some services	launchctl list
log show	Inspect system logs.	macOS alternative to journalctl	log show --last 1h

Quick Revision Set

- Use `pwd`, `ls`, `cd`, `mkdir`, `cp`, `mv`, and `rm` until moving around the file system feels automatic.
- Practice `cat`, `less`, `head`, `tail`, `grep`, `find`, and `diff` for reading and searching text.
- Use `chmod`, `chown`, `ps`, `top`, `kill`, `df`, `du`, `ping`, `ssh`, and `tar` as your core operating-system toolkit.
- Remember the common macOS replacements: `ifconfig` for `ip addr`, `diskutil` for `lsblk`, `vm_stat` for `free`, `launchctl` for `systemctl`, and `open` for `xdg-open`.